An Inherent Blind Spot of Physics-Based Attestation for Irrigation Logs, and Device-Anchored Repair in Monitoring, Reporting and Verification for Low-Emission Rice

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Abstract. Monitoring, reporting and verification (MRV) is the bottleneck keeping emission reductions in low-emission rice from becoming tradable carbon credits, and monitoring is about three quarters of that cost. Because every credit-eligible action under alternate wetting and drying is already an *actuation command*, a farmer-written irrigation log looks like free evidence of the highest rank under Decision 4801/QD-BNNMT. We built that layer on an FAO-56 digital twin, replayed logs under public ERA5 forcing, then attacked it with an optimising adversary. The paper's contribution is an inherent blind-spot proposition for every replay-based detector, plus a repair mechanism with near-zero marginal cost. The physics layer catches claimed depths beyond the soil's water-holding capacity, but an adversary who keeps the total volume and every individual depth and merely *shifts commands in time* inflates creditable dry phases by 33% in paired comparison. No physical tier escapes it usefully: the point model accuses 26.7% of honest farmers, and the robust tier that cuts false accusation to 5.0% in turn misses most time-shifting attacks. Over 281 audited logs at three Mekong Delta stations and two climate years, matching the log against the gateway's signed, time-stamped command record catches 116/116 forged logs and accuses none. Physics-based attestation cannot stand alone; the timestamp must come from a device, not the farmer.

Keywords: alternate wetting and drying; monitoring, reporting and verification; carbon credit; low-emission rice; digital twin.

1 Introduction

Viet Nam has committed to one million hectares of low-emission rice by 2030 [1], and the agronomy is settled: raising the severity of intermittent drying cut methane emission by 41–71% against continuous flooding with no grain-yield penalty [2], and water regime dominates seasonal methane flux across Asian rice fields [3]. What is not settled is measurement. Under Decision 4801/QD-BNNMT, documentary evidence for a credit claim is ranked, and the record of the irrigation and drainage actions actually taken sits at the top of that ranking: Form 3.3 of the spot-check in Annex 2 places actual irrigation and drainage records above plans, schedules,

field notes and signed statements [4], and under Verra's VM0051 the same record must survive third-party verification [5]. Producing it costs money a smallholder cooperative does not have: monitoring alone is about 77% of MRV expenditure [6], and the current procedure relies on periodic photographs of a water-level tube [4].

The obvious repair is to notice that a modern irrigation command is already a digital event: if a gateway actuates the pump it can also record the actuation, and that record is a Tier-1 activity datum in the sense of Decision 4801 [4] at no marginal cost — the premise of the companion work [7], which uses an FAO-56 twin as a safety gate deciding whether a command may be transmitted over a lossy channel. We took the next step and asked whether the same twin could *audit* what a farmer claims after the fact, with no sensor deployed: a log of claimed depths is replayed through the water balance under public ERA5 forcing [8], and a claim the soil could not have absorbed is evidence of fabrication. However, if the test field is generated by the very digital twin used to audit it — what we call *paper conditions* — the layer reports no false accusation on 60 honest logs and detection of 60/60 forged ones. Both results are correct but meaningless, because the trajectories labelled "real" were generated by the auditing model itself: the detector was grading its own homework (Supplementary S10.4).

This paper removes that circularity. We rebuild the test field so that it violates every assumption of the twin, and replace the naive fraud patterns with an adversary that has read the detector. Three findings follow, and they are also the contributions. First, the point-model detector is not merely optimistic, it is *unsafe*: it falsely accuses 26.7% of honest farmers, because an absolute pump-gap threshold treats a farmer who forgets one entry exactly like one who irrigated secretly; the robust rule that fixes this — an intersection over an uncertainty set Θ of 30 soil, crop-coefficient and efficiency configurations, each examined at all 5 initial states (Sec. 6.2) — pays for it in detection power, which we measure rather than hide. Second, physics-based attestation has a hole no amount of modelling closes: an adversary who preserves the total volume and every individual irrigation depth can *shift commands in time*, inflating the credit-eligible dry phases that VM0051 and Decision 4801 pay for by +33% in paired comparison ($t = 5.29$; +38%, $t = 3.46$, full uncertainty set). No physical tier escapes — the point model catches four of nine time-shifted logs but is the very detector that accuses 26.7% of honest farmers, while the robust tier that cuts false accusation to 5.0% lets eight of nine through — and we prove this is unavoidable for any detector that is a function of the replayed trajectory (Sec. 3, Sec. 5). Third, the repair is cheap and not physical: matching the log against the gateway's signed, time-stamped command record catches 116/116 genuinely forged logs and accuses none of 60 honest ones, at no cost beyond the gateway the companion work deploys (Sec. 7). The analysis also yields three direct consequences for the physical audit layer: the admissible depth is a range $[49.0, 86.0]$ mm, not 61 mm; the third physical filter cannot activate on dry-season delta rice; and the absolute pump-gap threshold is inadmissible (Sec. 6.1, Remark 1).

2 System Model

2.1 Field water balance

This paper reuses, unchanged, the standardised root-zone water balance of FAO-56 chapter 8 [9], so that every auditor starts from the same point. The root-zone water content w_t [mm] evolves hourly under the bucket equation:

$$w_{t+1} = \mathrm{clip}_{\le FC}\left(w_t + P_t - RO_t - ET_{c,t} + u_t\right), \tag{1}$$

That is, water in minus water out, with any excess above field capacity clipped away. Here:

w_t root-zone water content at hour t, in mm;

P_t rainfall during hour t, in mm;

RO_t surface runoff leaving the field during hour t, in mm, obtained from the curve-number method of the Soil Conservation Service (SCS-CN) published in 1972, evaluated daily and shared pro rata over the hours of a rain event;

$ET_{c,t}$ crop evapotranspiration during hour t, in mm;

u_t irrigation depth applied in hour t, in mm, the only term supplied by the farmer's log;

FC field capacity, the largest water content the soil retains after free drainage, in mm;

$\mathrm{clip}_{\le FC}$ the clipping operator, which passes values at or below FC unchanged and maps anything above it to FC.

$ET_{c,t}$ cannot be measured, so it is computed from reference evapotranspiration by FAO-56 equations 83 and 84:

$$ET_{c,t} = K_{s,t}\, K_c\, ET_{0,t}, \qquad K_{s,t} = \frac{FC - w_t}{(1-p)\,\mathrm{TAW}}, \tag{2}$$

where:

$ET_{0,t}$ hourly reference evapotranspiration, computed by the hourly form of Penman-Monteith, FAO-56 equation 53, from public weather data;

K_c the crop coefficient, characterising how much water rice uses relative to the reference grass;

$K_{s,t}$ the water-stress coefficient, falling from 1 towards 0 as the field dries, by FAO-56 equation 84;

p the fraction of available water the crop can still use before stress begins, taken from FAO-56;

TAW total available water in the root zone, equal to $FC - WP$ by FAO-56 equation 82.

Drainage above field capacity FC is removed by the clipping operator, so mass conservation holds exactly and is checked as an invariant.

What decides a dry phase is the relative depletion of the root zone, normalised to the interval from 0 to 1:

$$d_t = \frac{FC - w_t}{FC - WP}, \qquad \text{span} = FC - WP, \tag{3}$$

where:

d_t root-zone depletion at hour t, a dimensionless fraction equal to 0 at field capacity and to 1 at the wilting point;

WP the wilting point, the moisture below which the crop can no longer extract water, in mm;

span the width of the depletion scale, equal to $FC - WP$, used to convert the fraction d_t into a water amount in mm.

For the reference silty clay loam of the delta, the values $\mathrm{TAW} = FC - WP = 70\,\mathrm{mm}$, $FC = 160\,\mathrm{mm}$, $WP = 90\,\mathrm{mm}$ are the mid-range entries of FAO-56 Table 12 for a 0.5 m root zone [9]. These three equations are the whole physical basis the audit layer uses; no parameter is invented by us.

2.2 Replay of a claimed log

Let a claimed log supply the irrigation term of (1). The *replay operator* $\mathcal{R}$ maps a log to the trajectory it implies. The procedure substitutes the depths the farmer recorded into the u_t slot of (1), then iterates that equation hour by hour under the observed forcing $(P_t, ET_{0,t})$:

$$\mathcal{R}(L) = \{(\hat{w}_t, \hat{d}_t)\}_{t=0}^{T}, \tag{4}$$

with $\hat{w}_0 = w_{\text{init}}$ and

$$u_t = \sum_{(t_i, u_i) \in L} u_i \, \mathbb{1}[t_i = t], \tag{5}$$

where:

L the claimed log, a list of hour-and-depth pairs recorded by the farmer;

T the total number of hours in the simulation window;

$\hat{w}_t$ root-zone water content implied by the log, that is, the water the log *claims* the field had at hour t;

$\hat{d}_t$ the corresponding depletion fraction, obtained from $\hat{w}_t$ through (3);

$\hat{w}_0 = w_{\text{init}}$ the initial condition, the field's moisture state before the first irrigation;

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cdot]
)] the indicator function, equal to 1 when the bracketed condition holds and 0 otherwise.

The hat marks quantities reconstructed from the log rather than measured. This is the pivot of the whole paper: a replayed trajectory is what the farmer *declared*, not what the field produced. Replay is deterministic and uses only public reanalysis, so two auditors replaying the same log obtain the same trajectory and the verdict is reproducible. Fig. 1 summarises that path, from the farmer's log through the twin to the verdict of each defence tier. A superscript minus denotes the state immediately *before* that hour's claim is applied, as in $\hat{w}_t^-$: the water available to receive an event, which is what a filter must inspect.

The operator is also what makes the scheme cheap: no sensor is queried, and the only inputs are the log, the public weather series and a soil profile. Its weakness, developed in Sec. 5, is that it can only check whether a trajectory is *possible*, never whether it *happened*.

3 Threat Model

A claimed irrigation log is a finite map

$$L = \{(t_i, u_i)\}_{i=1}^{|L|}, \qquad u_i > 0, \; t_i \in [0, T), \tag{6}$$

where:

L the claimed log for one season, listing the irrigation events the farmer recorded;

$|L|$ the number of irrigation events in the log;

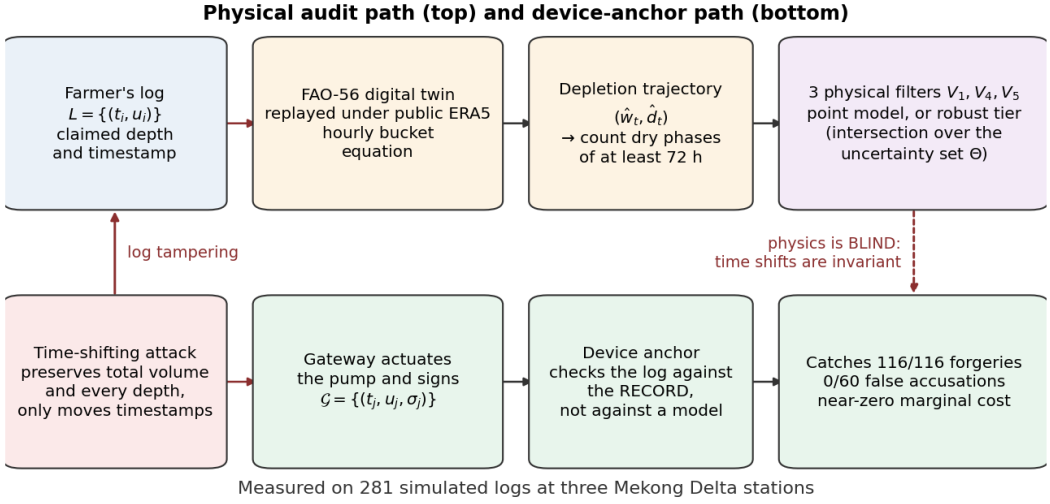


Figure 1. Data flow of the method. The physical audit layer replays the claimed log through the water-balance twin and then through three filters; the device anchor checks the same log against the signed command record of the gateway. The dashed arrow shows the time-shifting attack passing through the physical layer undetected.

u_i the claimed irrigation depth of event i, that is, the water the farmer records as applied to the field, in millimetres. This is the thickness of the water layer if that volume were spread evenly over the field surface, not a groundwater depth and not a soil-layer depth; it carries the same unit as the field capacity FC and the root-zone storage w_t defined in Sec. 2;

t_i the hour index of event i, taking values from 0 to T as defined in Sec. 2.

Three independent quantities can be forged, and each defence tier observes a different subset of them:

$$\underbrace{\textstyle\sum_i u_i}_{\text{total volume}} \quad , \qquad \underbrace{\{u_i\}}_{\text{individual irrigation depths}} \quad , \qquad \underbrace{\{t_i\}}_{\text{timestamps}} \quad . \tag{7}$$

Supplementary Table S1 lists the fraud classes corresponding to each. The adversary is *rational*: it maximises creditable dry phases subject to not being flagged, and it knows the detector, because the detector is fully specified in this paper and its source code is public; this is the sense in which ε in (12) below is not a secret: it is a published tolerance for model error, set to 5 mm throughout the experiments and equal to the value in the public source code. It is distinct from the pump-gap threshold τ_0 of Sec. 6.

The first defence tier compares the total claimed volume with the volume actually pumped. Let M be the volume actually pumped, measured at the cooperative station, and $U(L) = \sum_i u_i$ the claimed total. The tier-1 statistic is the gap

$$G = M - U(L), \tag{8}$$

where:

G the pump gap, in mm, being the volume really pumped in excess of the claimed total;

M the total volume actually pumped over the season, measured at the cooperative station, in mm;

$U(L)$ the sum of the depths claimed in log L, in mm.

A positive G means the farmer under-reported. The credit-relevant quantity, however, is not G but the number of *dry phases*. Write $\mathcal{A}(\cdot)$ for the set of maximal intervals $[a, b)$ of a replayed trajectory over which the depletion stays at or above d_{dry} with no claimed irrigation, lasting at least 72 h — the three-day minimum of Decision 4801 [4]:

$$\mathcal{A}(\{\hat{d}_t\}) = \Big\{[a,b) : \min_{t\in[a,b)} \hat{d}_t \geq d_{\text{dry}}, \quad u_t = 0 \ \forall t \in [a,b), \quad b - a \geq 72\,\text{h}\Big\}. \tag{9}$$

where:

$\mathcal{A}(\cdot)$ the operator returning the set of credit-eligible dry phases of a depletion trajectory;

$[a, b)$ a contiguous interval of hours, starting at hour a and ending before hour b, maximal in the sense that it cannot be extended further;

$\hat{d}_t$ depletion at hour t implied by the log, defined in Sec. 2;

d_{dry} the minimum depletion for an interval to count as a dry phase, a dimensionless fraction;

u_t the irrigation depth applied at hour t, equal to 0 throughout a dry interval;

$b - a$ the length of the interval, in hours.

The credited count is then

$$n_{\text{dry}}(L) = |\mathcal{A}(\mathcal{R}(L))|, \tag{10}$$

where $n_{\text{dry}}(L)$ is the number of credit-eligible dry phases of log L, $\mathcal{R}(L)$ is the replayed trajectory of that log under (4), and $|\cdot|$ counts the elements of a set. The threshold $d_{\text{dry}} = 0.45$ is used throughout; this value is a modelling choice of ours, set high enough that a dry phase is meaningful rather than merely not flooded (its sensitivity is examined in Appendix S20). Revenue is proportional to n_{dry}, not to $U(L)$ — the distinction that makes the attack of Sec. 5 profitable, invisible in a formulation that treats total volume as the target.

4 Physics-Based Filters and Their Exact Blind Spots

Replaying L under public ERA5 forcing produces $(\hat{w}_t, \hat{d}_t)$ (the companion work [7] gives the balance in full); three filters apply. V_1, *saturation*: a claim issued while the root zone is already at field capacity cannot be stored:

$$V_1(L) = \{\, i : \ u_i > 0 \ \wedge \ \hat{w}_{t_i}^{-} \ \geq \ FC - \varepsilon \,\}. \tag{11}$$

where $V_1(L)$ is the set of indices of the claims flagged by the first filter, and $\hat{w}_{t_i}^{-}$ is the root-zone water content immediately before event i is applied, defined in Sec. 2.

V_4, *storage excess*: a claim larger than the deficit available at that hour must have run off or percolated, so it was not "applied" in the sense the standard requires:

$$V_4(L) = \{\, i : \ u_i \ > \ \underbrace{(FC - \hat{w}_{t_i}^{-})}_{\text{deficit}} \ + \varepsilon \,\}. \tag{12}$$

where $V_4(L)$ is the set of indices of the flagged claims, and the underbraced expression is the deficit, that is, the water the root zone can still accept at that hour before reaching field capacity.

V_5, *infiltration bound.* Independent of state, a single event of duration τ_i cannot deliver more than the soil can accept:

$$u_i \;\le\; k_{\text{inf}}\,\tau_i + p_{\max}, \tag{13}$$

with $k_{\text{inf}} = 10\,\text{mm}\,\text{h}^{-1}$ and $p_{\max} = 100\,\text{mm}$ at the loose end of the published range for puddled rice [10, 11] (bound 110 mm at $\tau_i = 1\,\text{h}$), so that a farmer who records two adjacent events in one hour is not accused.

where:

τ_i the duration of irrigation event i, in hours;

k_{inf} the maximum infiltration rate of the field, in mm per hour;

$p_{\max}$ the maximum water the field accepts in one event regardless of duration, in mm.

A fourth filter, on depletion below the wilting point, was included in this layer. It cannot activate on dry-season delta rice: see Remark 1.

Proposition 1 (Exact blindness of aggregate defences)**.** *Let L and L' share the same multiset of depths and the same total, i.e. $\{u_i\} = \{u'_j\}$ as multisets. Then*

$$G(L) = G(L') \qquad \textit{and} \qquad |V_5(L)| = |V_5(L')|. \tag{14}$$

Proof. Supplementary S11; both quantities are symmetric functions of the depth multiset and never depend on t_i. □

Proposition 1 is the whole difficulty in one line: the two defences that do not depend on the model being right — the pump meter and the infiltration bound — are *exactly* invariant under the transformation the adversary needs. What remains is V_1 and V_4, which do depend on order through $\hat{w}^-_{t_i}$; they turn out not to help.

5 The Time-Shifting Attack

Proposition 2 (Credit gain and evasion are the same action)**.** *Let a claim at hour t have available deficit $c_t = FC - \hat{w}^-_t$. Between consecutive claims the bucket loses $\Delta_t = \sum_s (ETc_s + DP_s - P^+_s) \ge 0$, so delaying a claim by δ hours increases its deficit by the drainage accrued in those hours:*

$$c_{t+\delta} \;\ge\; c_t + \min\!\big(\delta \cdot \overline{\Delta},\; FC - \hat{w}^-_t\big). \tag{15}$$

where:

c_t *the deficit at hour t, in mm, equal to the water the root zone can still accept before reaching field capacity;*

Δ_t *the water the bucket loses between two consecutive claims, in mm;*

ETc_s *crop evapotranspiration at hour s, that is, $ET_{c,s}$ of* (2)*;*

DP_s *deep percolation below the root zone at hour s, the water removed by the clipping operator in* (1)*;*

P_s^+ *the rainfall actually entering the field at hour* s*, after surface runoff* RO_s *has been subtracted;*

δ *the number of hours by which a claim is delayed;*

$\overline{\Delta}$ *the mean water loss per hour over the waiting interval.*

All three terms of Δ_t *come from the same bucket equation* (1)*, so this inequality rests on no new hydrological assumption. When a command is recorded later than it was issued, the field dries further during the wait, so the number of dry phases rises, while the drier soil absorbs exactly that water and no physical filter sees anything anomalous.*

Consequently the transformation that moves claims out of *a dry interval and onto its boundary simultaneously* (*i*) *increases* n_{dry}*, because the interval is no longer interrupted, and* (*ii*) *increases* c_t *at every moved claim, which by* (12) *makes* V_4 less *likely to fire. There is no trade-off for the adversary to negotiate: the gradient of profit and the gradient of undetectability point the same way.*

Proof. Supplementary S11. In short: (i) is definitional, since $\mathcal{A}$ requires no claim inside the interval; (ii) follows because delaying a claim lets the bucket drain for δ further hours with no inflow, so the available deficit can only grow while u_i is unchanged. □

This is why the attack cannot be closed by better hydrology: any detector whose verdict is a function of the replayed trajectory must integrate the same drainage that makes the delayed claim look legitimate. The empirical attack is a hill-climb over integer hour offsets in $[-36, 36]$ accepting a move only if the total is preserved bit-for-bit, every depth stays under (13), and n_{dry} strictly increases. The attack does not steal water — what is stolen is *history* (Supplementary S19).

6 When May a Farmer Be Accused?

6.1 The admissible depth is a range, not a number

This layer stated a single bound, $u \leq 61\,\mathrm{mm}$, as the largest claim that never raises an alarm — an artefact of one soil profile and one initial condition. With V_4 firing when $u_i > c_{t_i} + \varepsilon$ and $c_t = d_t \cdot \mathrm{span}$, the bound at a field with initial depletion w_0 irrigating no later than d^* is

$$u_{\max} = \max(d^*, w_0) \cdot \mathrm{span} + \varepsilon. \tag{16}$$

where:

$u_{\max}$ the largest claimed depth, in mm, that never raises an alarm;

d^* the depletion at the latest irrigated hour considered, a dimensionless fraction;

w_0 the initial depletion of the field, in the same fraction units;

span the width of the depletion scale, equal to $FC - WP$ in mm, defined in Sec. 2;

ε the published tolerance for model error, equal to 5 mm as in Sec. 4.

This bound is written in depletion relative to the root zone rather than against an absolute elevation, so unlike the absolute water-table thresholds used in current practice it does not drift as

the field surface subsides; the proposition and the measured shift for the delta are in Supplementary S4.

Two corrections follow, both verified by scanning the replay operator rather than by algebra alone (derivation in Supplementary S14): (16) uses max, not a difference — an initially dry field admits *larger* claims than a wet one, and the difference-form expression corresponds to no physical state — and it bounds a *single event*, not the season, so an adversary issuing many sub-threshold claims can inflate the total without bound. Supplementary Table S2 gives the range that must be reported instead, $[49.0, 86.0]$ mm across Θ.

6.2 Robust accusation under model uncertainty

The auditor knows neither the farmer's soil, nor the growth stage, nor the delivery efficiency. Let Θ be the uncertainty set and $V_\theta(L)$ the violation set under configuration θ; the point-model detector evaluated a single default θ_0 and treated every violation as guilt. We replace it by an intersection:

$$\begin{aligned} V_{\mathrm{rob}}(L) &= \bigcap_{\theta \in \Theta} V_\theta(L), \\ \Theta &= \mathcal{S} \times \mathcal{K} \times \mathcal{E} \times \mathcal{W}, \end{aligned} \tag{17}$$

where:

$V_{\mathrm{rob}}(L)$ the robust violation set of log L, non-empty only if L violates under every configuration;

Θ the uncertainty set, the product of the four component sets below;

$\mathcal{S}$ the set of soil textures, of size 5;

$\mathcal{K}$ the set of crop-coefficient regimes, of size 2;

$\mathcal{E}$ the set of irrigation efficiencies, of size 3;

$\mathcal{W}$ the set of initial states, of size 5.

Hence $|\Theta| = 150$ and each log costs 150 replays. The semantics are deliberately one-sided:

$$a(L) = \begin{cases} \text{accuse}, & V_{\mathrm{rob}}(L) \neq \emptyset, \\ \text{query}, & \text{otherwise.} \end{cases} \tag{18}$$

where $a(L)$ is the verdict for log L. A "query" verdict means further field verification is required; the system does not conclude fraud on its own.

Proposition 3 (No false accusation within Θ)**.** *If the true field configuration θ^* belongs to Θ and L is compliant under θ^*, then $V_{\mathrm{rob}}(L) = \emptyset$ and the farmer is not accused.*

Proof. $V_{\mathrm{rob}}(L) \subseteq V_{\theta^*}(L) = \emptyset$. □

Proposition 3 is what makes the layer deployable: model error can reduce detection power but cannot manufacture an accusation, at a price that must be stated. Two implementation details — the treatment of non-firing worlds and the span of $\mathcal{W}$ — silently break the proposition if got wrong (Supplementary S7).

6.3 A proportional pump-gap threshold

The absolute rule $G > \tau_0$ with $\tau_0 = 5\,\text{mm}$ assumes the log is a faithful copy of what was pumped. It is not: forgetting, rounding to the nearest 5 mm and day-late entries all contribute to G without wrongdoing, so the threshold must scale with the number of recorded events:

$$\begin{aligned} \tau(n) &= \delta_{\text{rel}} M + z\sqrt{n}\,\sigma_{\text{event}}, \\ \sigma_{\text{event}}^2 &= p(1-p)\bar{u}^2 + \tfrac{r^2}{12} + q(1-q)(s\bar{u})^2, \end{aligned} \tag{19}$$

where:

$\tau(n)$ the admissible pump gap, in mm, when the log records n events;

n the number of irrigation events recorded in the log;

δ_{rel} the relative tolerance on the true pumped volume M, equal to 2%;

z the multiplier of the standard deviation, set to 3;

σ_{event} the standard deviation of the recording error per event, in mm;

p the rate of forgetting to record an event;

r the depth rounding step, equal to 5 mm;

q the depth-error rate, and s its relative amplitude;

$\bar{u}$ a typical event depth, equal to 30 mm.

Here the absolute rule is the degenerate case $n \to 0$, $\sigma_{\text{event}} \to 0$.

Remark 1 (A filter that cannot fire)**.** *The third physical filter, flagging depletion below the wilting point, was doubly inoperative: its threshold* (1.02) *lay outside the clipped range of the reported indicator* (0 *firings in* 281 *logs*)*, and once repaired it still cannot activate — across* 150 *configurations with an empty log the unclipped depletion peaks at* 0.9987*, and a dry-season delta rice field does not reach the wilting point* (*Supplementary Table S3, derivation in S13*)*. The layer has two operative physical filters and one inert safety device.*

7 Device-Anchored Attestation

7.1 The gateway record

Propositions 1–2 show that volume and depth are insufficient and that order is exactly what the adversary manipulates, so the defence must observe timestamps — and the only party that can vouch for a timestamp is the device that acted on it. Let the gateway hold a signed command record

$$\mathcal{G} = \{(t_j, u_j, \sigma_j)\}, \tag{20}$$

where:

$\mathcal{G}$ the gateway's command record, covering every pump actuation in the season;

t_j the hour at which the controller switched the pump for command j;

u_j the water depth the controller delivered for that command, in mm;

σ_j a digital signature over the pair (t_j, u_j), produced by the controller at actuation time.

The anchor test compares the claimed log against it:

$$\begin{aligned} \Delta_{\text{time}} &= \{t_i\} \triangle \{t_j\}, \\ \Delta_{\text{depth}} &= \max_t |u(t) - u_{\mathcal{G}}(t)|, \end{aligned} \tag{21}$$

where:

Δ_{time} the set of hours present in one record but absent from the other;

$\triangle$ the symmetric difference of the two hour sets, returning elements belonging to only one of them;

$u(t)$ the depth claimed at hour t in log L;

$u_{\mathcal{G}}(t)$ the depth the gateway recorded at the same hour t;

Δ_{depth} the largest discrepancy between the two depths, taken over hours common to both records.

The verdict is then

$$A(L, \mathcal{G}) = \begin{cases} \text{forge}, & \Delta_{\text{time}} \neq \emptyset \ \vee \ \Delta_{\text{depth}} > \epsilon, \\ \text{match}, & \text{otherwise}, \end{cases} \tag{22}$$

where:

$A(L, \mathcal{G})$ the verdict of the anchor test on log L;

ϵ the depth-matching tolerance, equal to 10^{-6} mm, i.e. exact agreement up to floating-point noise;

$\vee$ logical or, meaning that either discrepancy alone suffices to conclude forgery.

Naming the two discrepancies separately matters: time-shifting alters Δ_{time} only, depth inflation alters Δ_{depth} only, and omission and addition alter both, so a single comparison catches every fraud class.

Proposition 4 (Completeness of the anchor)**.** *If a claimed log L differs from the gateway record $\mathcal{G}$ in the multiset of hours or in any depth at a shared hour, then $A(L, \mathcal{G})$ returns forge. If L and $\mathcal{G}$ agree on both, then L is a faithful transcript and no forgery of the three quantities in* (7) *has occurred.*

Proof. Supplementary S12: the first clause is the definition of (22), and the second follows because all three forgeable quantities are functions of the hour set and the per-hour depths. □

7.2 Why this does not re-introduce false accusation

Proposition 4 is strong enough to look dangerous: any discrepancy is treated as forgery, including the forgetting and rounding that Sec. 6 showed to be ubiquitous. The resolution is that the anchor compares a transcript against the *source*, not a log against a *model*: an honest farmer who forgets an entry is simply completed from $\mathcal{G}$. False accusation then requires the gateway to disagree with itself — a device-integrity question — which is why Table 1 holds at 0/60 while the behavioural tiers hold at 26.7% and 5.0% (Supplementary S18).

7.3 Cost and the scope of the claim

The anchor requires the gateway of the companion work [7], which actuates the pump and already writes a signed, time-stamped command record as part of its safety function; the marginal cost of attestation is storage and transmission of a record that exists regardless (Supplementary Table S4). The scope is nonetheless narrower than the premise we started from: a hand-written log, with no device in the loop, is *not* admissible evidence on its own, since Sec. 9.2 measures a rational adversary inflating credits by +35% while escaping every physics- and volume-based test (6/6 altered logs). Near-zero-*marginal*-cost MRV is achievable only where a controller stands between the farmer and the pump; where irrigation is manual this layer does not apply.

8 Experimental Design

The design compares four defence layers against nine log types on a field that deliberately violates every assumption of the twin: if the field were generated by the twin itself, a 0/60 false-accusation rate would measure only self-consistency (Supplementary S10.4), so here the field is generated by a separate process that differs from the twin along seven axes, each with a published source (every value in Supplementary Table S5; axis-by-axis list and per-event recording semantics in S20.1). Weather is hourly ERA5 [8] (Open-Meteo archive) for Can Tho, Soc Trang and Ca Mau over the water-scarce year 2024 and the water-surplus year 2025, window 1 March to 29 May (2 160 h), the dry season in which AWD is practised [12]. Layer 1 is the point-model detector (default profile, absolute threshold $\tau_0 = 5\,\mathrm{mm}$); Layer 2 the robust detector of (17) with the proportional threshold of (19), on both a reduced Θ (24 replays per log, 10 seeds) and the full one (150 replays, cross-check); Layer 3 the device anchor of (22); Layer 4 physics alone, for reference. Of the nine log types, two are honest (paper and field conditions), three are naive forgeries and four adversarial, built by an optimiser rather than a fixed recipe (full list in S16); all attack statistics are conditioned on a `changed` flag, because an adversary that finds no improving move returns the original log, and comparisons use the baseline of the variant itself (two harness defects, S10.2). For each layer we report the escape rate (forged logs neither physical tier flags) and the false-accusation rate on honest logs; for credit impact, n_{dry} of (10) against the honest field baseline, paired. Volume appears only as a diagnostic (S10.3).

9 Results

9.1 The point-model detector is unsafe, not merely optimistic

Table 1 gives the central negative result: the point-model detector accuses nobody on paper-condition logs, reproducing the 0/60 of the self-consistency protocol, but accuses 26.7% of honest farmers on field-condition logs. The cause is (8) with an absolute threshold — forgetting, rounding or a day-late entry all produce $G > 5\,\mathrm{mm}$ without wrongdoing; the measured gap on honest field-condition logs reached 65 mm. The proportional threshold of (19) cuts this to 5.0%, the device anchor to 0/60.

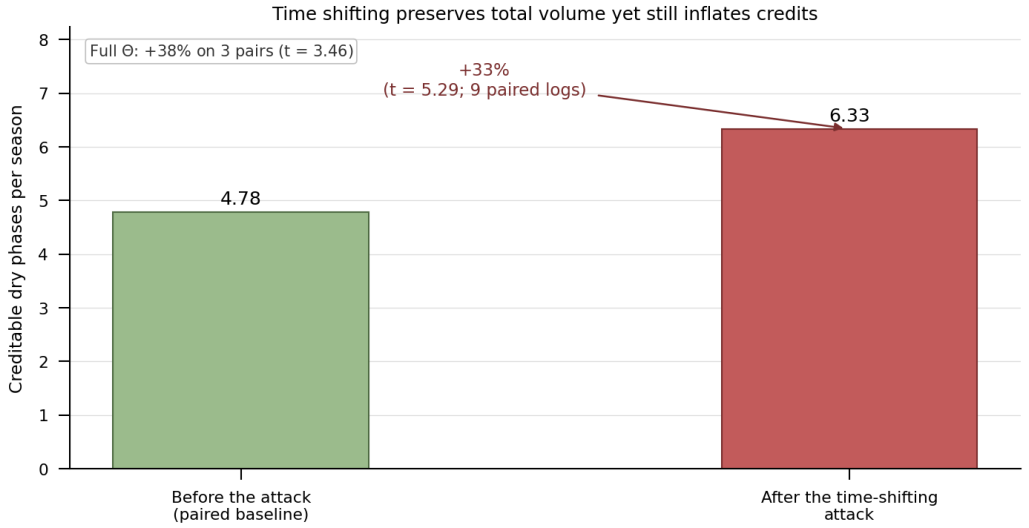


Figure 2. The time-shifting attack on the paired subset. Creditable dry phases before and after moving the command timestamps, with the total volume and every individual depth preserved. The gain under the full uncertainty set is noted at the lower right.

9.2 Physics cannot see the attack that pays

Fig. 2 plots that paired comparison. Table 2 is the second negative result, and the one that matters economically. Every inflation figure is a *paired* comparison against the honest log of the same trajectory, restricted to attacks that actually changed the record; the unpaired alternative overstates the gain roughly twofold (S15). Time-shifting raises creditable dry phases from 4.78 to 6.33 per season, +33% over nine altered logs ($t = 5.29$); under the full uncertainty set, +38% over three ($t = 3.46$, $p = 0.074$ — directionally consistent but not significant at 0.05 with this sample); the rational adversary reaches +35% and +40% ($p = 0.007$ and $p = 0.30$). Consolidation gains almost nothing once paired (+1%, +2%), and the two crude forgeries *reduce* creditable phases (−1%, −17%): only the volume-preserving permutation of timestamps is profitable, as Proposition 2 predicts. The escape rates in Table 3 are the heart of the result: time-shifting escapes the robust physics layer in 8 of 9 altered logs, the rational adversary in 6 of 6. The point model catches more (4 of 9), but it is precisely the detector that accuses 26.7% of honest farmers; the robust tier cuts that to 5.0% and hands most of the detection back (1 of 9) — *no physical configuration escapes that trade-off*, because both tiers are functions of the replayed trajectory. The infiltration filter V_5 fires 648 times, but exclusively on the crude depth-inflation forgery: useful, and orthogonal to the threat that pays.

9.3 The anchor closes every class, and the cost of robustness

Fig. 3 summarises the trade-off measured in the two preceding subsections. Table 3 also gives the constructive half. All 281 logs were audited under the reduced uncertainty set, of which 85 were re-audited under the full one (Supplementary S17.2). Of the 281 logs, 116 were genuinely altered, and the device anchor caught 116/116 while accusing none of 60 honest field-condition logs (37/37

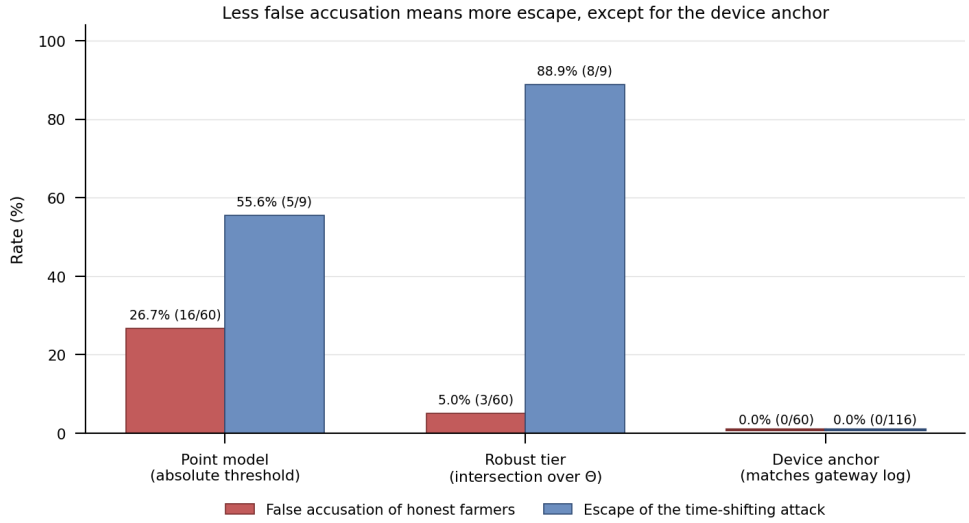


Figure 3. Trade-off between the two error types across the three defence tiers. Lowering the false-accusation rate raises the escape rate of the time-shifting attack; only the device anchor reaches zero on both.

Table 1. False accusation on honest logs, the property the scheme is sold on. Paper conditions reproduce the 0/60 of the self-consistency protocol; field conditions expose the absolute gap threshold.

Defence layer	Honest, paper conditions	Honest, field conditions
Point model, physics only	0/60	2/60
Point model with absolute threshold ($G > 5\,\mathrm{mm}$)	0/60	16/60
Robust rule ($\forall$ over Θ), physics only	0/60	0/60
Robust rule with proportional $\tau(n)$	0/60	3/60
Device-anchored command log	0/60	0/60

and 0/18 under the full Θ). Naive forgeries are caught by every layer; the distinction lies in the adversarial rows, where only the anchor holds. Robustness is not free: on the nine omission-fraud logs actually altered the point model catches 9/9, the robust rule 4/9 (4/4 vs 2/4 full Θ) — a loss of detection power, not a Pareto improvement, though the correct side of the trade. Every conclusion holds under both the reduced Θ (24 replays, 281 logs, 10 seeds) and the full one (150 replays, 85 logs, 3 seeds): false accusation 0/60 and 0/18, anchor completeness 116/116 and 37/37, time-shift escape 8/9 and 3/3, paired inflation +33% and +38% (Supplementary S17).

10 Discussion

10.1 Relation to prior work

AWD monitoring has been approached from three directions — instrumented water-level sensors [15], remote sensing [16–19], and programme-level MRV [1, 20, 21] — while agronomic meta-

Table 2. Creditable dry phases of at least 72 h, the three-day minimum without re-flooding set by Decision 4801 and the quantity VM0051 credits against, versus the honest field baseline, paired. The last column counts attacks that actually altered the log. The table carries two baselines, deliberately different: the inflation column uses the paired baselines against the honest log of the same station, year, seed and soil (4.87 for the adversarial variants), while the "honest" row reports the mean 3.23 phases over all 60 honest logs. See Supplementary S15.

Log type	Altered logs	Paired dry phases (before → after)	Inflation	*t*
Honest farmer (field conditions)	—	3.23 (reference)	reference	—
Added irrigation events	23/23	4.87 → 4.83	-1%	-0.57
Infeasible claimed depth	23/23	4.87 → 4.04	-17%	-4.03
Omission of real commands	9/23	5.44 → 4.78	-12%	-2.83
Claim consolidation, 24 h	23/23	4.87 → 4.91	+1%	0.37
Claim consolidation, 96 h	23/23	4.87 → 4.96	+2%	0.62
Time shifting (rational)	9/23	4.78 → 6.33	+33%	5.29
Rational adversary	6/23	4.33 → 5.83	+35%	4.39

Table 3. Main result. Rates are computed over the logs an attack *actually altered*; column 2 reports altered over generated, since the adversary returns the original log when it finds no improving move. Escape rate is the fraction neither physical tier flags.

Fraud variant	Altered logs	Escapes point model	Escapes robust detector	Caught by anchor
Honest farmer (field conditions)	0/60 (honest)	**27%** accused	5% accused	0% accused
Added irrigation events	23/23	0%	0%	23/23
Infeasible claimed depth	23/23	0%	0%	23/23
Omission of real commands	9/23	0%	56%	9/9
Claim consolidation, 24 h	23/23	39%	91%	23/23
Claim consolidation, 96 h	23/23	39%	91%	23/23
Time shifting (rational)	9/23	56%	89%	9/9
Rational adversary	6/23	100%	100%	6/6
All genuinely forged	**116**	**—**	**—**	**116/116**

Table 4. Confirmable credit-eligible dry phases under each MRV route, over the 150 phases present in the 60 simulated fields; a phase counts only if at least one observation falls inside it. The final two rows tie here and differ in cost and in what they prove about a doctored log.

Verification route	Confirmable dry phases ($n = 150$)	Observations per season	Devices per plot
Photo + self-report (current practice) [4]	0/150 (0%)	—	0 (PVC tube)
Sentinel-1A+1B (6-day revisit) [13]	124/150 (83%)	15	0
Sentinel-1A alone (12-day revisit) [14]	90/150 (60%)	8	0
Continuous water-level tube (IoT) [15]	150/150 (100%)	2,153	1 per plot
Signed command record (this work) [7]	150/150 (100%)	2,153	0 (controller)

analysis [22] and adoption studies [23, 24] explain why instrumented monitoring has not spread. None evaluates the *reliability of the action record itself*, which is the question asked here (full survey in Supplementary S8). Satellite radar, the obvious low-cost alternative, we measured rather than argued about: Table 4 scores every route by how many of the 150 credit-eligible dry phases actually present across the 60 simulated fields it could confirm (Supplementary S9.2), a phase counting only if at least one observation falls inside it [13–15]. Two cautions: the column measures the ability to *earn* a credit, not to detect fraud — tube and command record tie at 100% but differ in cost and in what they prove — and photograph-plus-self-report scores 0% because it produces no observation series to reconcile against. Radar therefore keeps the role VM0051 Appendix 4 assigns it: a per-plot cross-check lowering the frequency of field verification [5] (Supplementary S9.2).

10.2 Limitations

Seven limitations bound the claims, all stated in full in Supplementary S21; the two that condition the headline results are these. *Simulated farmers*: forgeries are injected into simulated trajectories, so a field pilot is still needed to measure behavioural response — above all whether farmers shift their *actual* pumping once the anchor is known, an attack that changes M rather than L and that our model cannot represent. *The attack is demonstrated in the water-scarce year only*: in 2025 most logs fall below the three-command minimum at which a timing attack is constructible — of 37 genuinely altered logs under the full Θ, all 37 come from 2024. That escarpment is itself a finding — MRV fraud risk is concentrated where water is scarce — but the attack results are not yet evidence about a wet season; under the reduced Θ twelve altered logs do come from 2025 and behave the same way — suggestive rather than conclusive. The remaining five are enumerated in S21.

11 Conclusion

We set out to make rice MRV nearly free by auditing a farmer-written irrigation log against soil-water physics. The layer reported no false accusation and complete detection; both numbers were

artefacts of a field generated by the auditing model itself, and rebuilding it against an adversary that optimises the credited quantity changes the conclusion.

The point-model detector falsely accuses 26.7% of honest farmers; a proportional threshold and an intersection over a 150-element uncertainty set reduce that to 5.0% and 0/60 at a measured cost in detection (9/9 to 4/9 on omission fraud). Meanwhile a volume-preserving permutation of claimed times inflates creditable dry phases by 33% (paired, $p < 0.001$; 38% under the full uncertainty set, but on three logs only and thus not significant) while escaping the robust tier in 8 of 9 and 3 of 3 altered logs, and no physical configuration escapes that trade-off: delaying a claim both lengthens the dry phase and increases the deficit available to it, so profit and undetectability share a gradient.

The repair is not hydrological. Matching the claimed log against the gateway's signed, time-stamped command record catches 116/116 genuinely forged logs and accuses none of 60 honest ones, at no cost beyond a controller the companion work already deploys. What survives of the original premise: physics *constrains* a log, but only a signed device record *attests* it — near-zero-*marginal*-cost MRV is achievable only where a device stands between the farmer and the pump. Three practical consequences for the physics-audit layer, already stated in the Introduction, are recalled here: the admissible depth is a range $[49.0, 86.0]$ mm rather than 61 mm; the third filter is inert; and an absolute gap test is inadmissible. Code, both uncertainty sets, all 281 audited logs and 46 unit tests are public.

Data, Code and Declarations

Code, both uncertainty sets, both adversaries and all 281 audited logs are at `https://github.com/mxuanvan02/TwinGate` under the MIT licence; weather forcing is public ERA5 reanalysis via Open-Meteo. Funded by Hue University, Grant No. DHH2025-19-07. No competing interests. Ethics approval is not applicable (simulation and public data only).

Tóm tắt và từ khóa (Tiếng Việt)

Đo đạc, báo cáo, thẩm định (MRV) là nút thắt khiến phần giảm phát thải của lúa phát thải thấp chưa thành tín chỉ các-bon, vì giám sát chiếm khoảng ba phần tư chi phí. Trong tưới khô ướt luân phiên (AWD), mỗi hành động đủ điều kiện cấp tín chỉ là một lệnh bật máy bơm, nên nhật ký nông dân tự ghi trông như bằng chứng miễn phí, xếp ở bậc ưu tiên cao nhất theo Quyết định 4801/QĐ-BNNMT. Bài báo dựng lớp kiểm toán đó trên bản sao số (digital twin) FAO-56, phát lại nhật ký với ERA5 công khai rồi cho bên gian lận tối ưu hoá tấn công. Kết quả, mọi bộ phát hiện dựa trên quỹ đạo phát lại có điểm mù tất yếu, kèm cơ chế khắc phục có chi phí biên gần bằng không. Tầng vật lý phát hiện khai báo vượt sức chứa ẩm đồng ruộng, nhưng bỏ lọt bên gian lận dịch thời điểm lệnh, giữ nguyên tổng lượng, khiến số pha khô được cấp tín chỉ tăng 33% (ghép cặp). Bộ phát hiện mô hình điểm kết tội oan 26,7% nông dân trung thực; bộ bền vững hạ oan còn 5,0% nhưng lọt phần lớn tấn công dịch thời điểm. Trên 281 nhật ký ở ba trạm Đồng bằng sông Cửu Long, phép neo bắt 116/116 ca giả mạo, không oan ai. Vật lý do đó chỉ ràng buộc nhật ký; dấu thời gian phải

do thiết bị cấp.

Từ khóa: tưới khô ướt luân phiên (AWD), đo đạc báo cáo thẩm định (MRV), tín chỉ các-bon, lúa phát thải thấp, bản sao số.

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